Implementation of a Dynamic Traffic Light System Using Fuzzy Logic in SUMO Simulation

Alan Nabiel
Department of Computer Science
Faculty of Computer Science, Bina Nusantara University
Alam Sutera, Tangerang
alan.nabiel@binus.ac.id

Ian Frederick
Department of Computer Science
Faculty of Computer Science, Bina Nusantara University
Alam Sutera, Tangerang
ian.hartanto@binus.ac.id

Ivan Sebastian Edbert S.Kom., M.Kom.
Department of Computer Science
Faculty of Computer Science, Bina Nusantara University
Alam Sutera, Tangerang
ivan.edbert@binus.ac.id

Alvina Aulia S.Kom., M.TI.
Department of Computer Science
Faculty of Computer Science, Bina Nusantara University
Alam Sutera, Tangerang
aaulia@binus.edu

Abstract*— Conventional fixed-schedule traffic signals significantly aggravate urban congestion, leading to extended vehicle idling, increased fuel use, economic burdens, and environmental harm. This study tackles these limitations by designing, implementing, and appraising an intelligent control sytstem traffic system powered with fuzzy logic, utilizing Simulation of Urban Mobility (SUMO) environment. The developed fuzzy logic controller (FLC) responsively alters signal timings by analyzing real-time SUMO data, specifically the count of vehicles on lanes on green and also red light, and the recorded time wait at red light signals. A rigorous comparative analysis assessed the FLC against a traditional fixed-time controller (FTC) through simulations of an isolated four-approach intersection, with individual simulation runs lasting for approximately 16,000 steps. The findings reveal the FLC's pronounced advantages, delivering an estimated 25.2% decrease in mean vehicle waiting duration, a 7.0% reduction in the aggregate number of vehicle stops, a 27.2% drop in cumulative network delay, and a 6.0% rise in average junction throughput. Such results confirm that fuzzy logic-based approaches hold considerable promise for substantially improving traffic flow smoothness, reducing traffic jams, and enhancing city-wide mobility.

Keywords*— Dynamic Control, Fuzzy Logic, SUMO simulation, and Traffic Optimization

I. INTRODUCTION

Standard traffic light systems function with predetermined cycles that don't adjust to live traffic flow, often leading to extended wait times, wasteful fuel use, and significant pollution, which consequently have adverse effects on the environment and economy.[1].

As urban traffic congestion increases, the need for adaptive traffic control becomes increasingly important. According to the Urban Mobility Report 2019, in 2017 the total cost of congestion around United States Urban area was $179 billion and the cost is increasing every year [2] . In Indonesia, traffic congestion in Jakarta alone has caused significant economic losses. A study cited in [3] , estimated that traffic-related economic losses, in terms of lost time, fuel costs, and health impacts, reached Rp. 12.8 trillion per year in 2010 and is projected to increase to Rp. 65 trillion by 2020, equivalent to approximately €4.13 billion. To address the current mobility degradation, one promising solution is to implement dynamic traffic light control, which can optimize traffic flow by making real-time decisions based on actual traffic conditions[4].

Unlike static traffic lights where the rules are completely inadaptive to the field conditions that occur, Static Fuzzy Logic provides flexibility so that it can make adaptive decisions that will make traffic more efficient. This system

provides superior performance compared to traditional systems because the capability to reduce vehicle stops and shorten the length of time queues, especially in heavy traffic conditions[5]. Static Fuzzy Logic implemented by the scikit-fuzzy library can consider the duration of the traffic light even though it still has pre-defined fuzzy rules. Although the fuzzy rules are predefined and not self-learning, system's ability able to process uncertain and imprecise traffic data, such as fluctuating vehicle counts or unpredictable congestion patterns,makes it well-suited for application in dynamic traffic light management[6].

To test the effectiveness of the dynamic traffic light system, this study uses the Urban Mobility Simulation (SUMO) as a trial platform [7]. SUMO can model each vehicle individually with certain characteristics that can be specifically set so that the simulation can better imitate traffic conditions[8]. According to Behrisch et al. (2011), SUMO is able to calculate exhaust emissions and fuel consumption using a data-based emission model from HBEFA so it is very suitable for use as a traffic light simulation, while calculating vehicle fuel efficiency [9] . SUMO also offers a cost-effective testing environment that facilitates extensive experiments without the need for physical infrastructure, making it an ideal platform for hypothesis validation before real-world implementation.

Several previous studies have explored dynamic traffic light systems using different methodologies. For example:

• POP-TRAFFIC: Road Traffic Analysis and Prediction using Fuzzy Neural for Approach [10] developed a dynamic traffic control system using Fuzzy Neural Network models, specifically Pseudo Outer Product Based Fuzzy Neural Network with Variant Time Reinforcement. This system processes data from field sensors and then validates it through video recordings.

• Smart Traffic Light Control System proposed an adaptive traffic light system that utilizes magnetic, infrared, and camera sensors to be able self-adjusting based on real-time data of traffic [11].

Although these studies have proven their advantages compared to static traffic systems, in their implementation, these studies still require physical sensors or devices that limit flexibility and scalability and will add costs to expensive physical infrastructure. Our research focuses on integrating Fuzzy Logic adaptive control traffic light running on SUMO environment, which allows for extensive experimentation, validation, and optimization in a controlled simulation environment before implementation in real-world settings. By simulating various traffic conditions in SUMO, we intend to analyze how the Fuzzy Logic-based system compares with the traditional fixed-time traffic light control method. So that before carrying out large-scale implementation, the feasibility and effectiveness of this Fuzzy Logic-based traffic system can be tested and understood.

II. LITERATURE REVIEW

2.1 Fuzzy Logic in Traffic Light Control

Fuzzy logic is one of the most effective approaches to handle traffic conditions that are always changing and unpredictable. Different approaches and data sources can also apply fuzzy logic in their research.

The paper titled "*Dynamic Traffic Light Controller Using Machine Vision and Optimization Algorithms*" [12] uses Fuzzy Logic in a centralized system consisting of three main modules. The first is Next phase to determine which light phase should be activated immediately based by the number data of vehicle, also how long the vehicle has been delayed , then continued with Observation to monitor the current green phase, whether it needs to be stopped or not. The last is Decision to decide on the phase change, so this system is quite effective in forming a green wave through coordination between intersections and finally shows a reduction in waiting time of 20% -26%

Meanwhile, in the paper by Arena et al. [13] , the system uses several integrated Fuzzy Logic Controllers (FLC) by obtaining data via communication between vehicles (V2V / V2I) with the IEEE 802.11p and LTE-V2V protocols. Each Controller can adjust to traffic conditions with emergency vehicle priority. This system not only responds to the number of vehicles but also their types because it can prioritize emergency vehicles.

In the last paper that we analyzed, Collotta et al. [14], Fuzzy Logic is applied in the form of four parallel fuzzy controllers, where each handles one phase of the traffic light at an intersection that works independently without considering the traffic conditions of other intersections. Each controller calculates thegreen light length based on the the two-lane queue length related to the input data obtained using an Wireless Sensor Network that can monitor many vehicles in real time.

From the three papers, it shows that Fuzzy Logic can be used for traffic systems with various types of supporting infrastructure as needed. The first paper is more suitable for areas with camera systems and if it requires coordination of traffic light settings at several adjacent intersections. Then the second paper is very suitable for urban environments that have V2X support, and the third paper is the most likely to implement economical dynamic lights at isolated intersections.

2.2 SUMO as a Simulation Platform

After discussing and analyzing the fuzzy logic approach to traffic management, it is very important to test and evaluate these systems in a simulation environment, one of which is an open sourced software called SUMO or Simulation of Urban Mobility. SUMO is used for conducting traffic simulations and is often used to evaluate dynamic traffic control systems.

Because it can be used as a simulation platform with different purposes, SUMO is a fairly flexible simulation platform and can provide a cost-effective solution for testing the effectiveness of dynamic traffic light systems. Like Faye

et al. [15]who used SUMO to evaluate distributed algorithms to control traffic at isolated intersections with a focus on reducing waiting times. On the other hand, Lou and Vorobeychik (2016) in the paper titled "*Decentralization and Security in Dynamic Traffic Light Control*"[16] utilized the SUMO platform with a focus on investigating aspects of decentralization and prioritizing security in dynamic traffic light control and adding complexity through game theory. Finally, Hossan and Nower [17] used SUMO to test a distributed architecture that uses a fog computing-based system with a focus on reducing waiting times and increasing fuel efficiency.

All three papers use SUMO as a simulation platform to test the feasibility of their systems while exploring various aspects. SUMO can be used to model various complex traffic scenarios and can support complex control system simulations, proving that SUMO can be relied on as a cost-effective form of evaluation.

After analyzing the fuzzy logic approach to traffic management, it is very important to test and evaluate these systems in a simulation environment, one of which is SUMO.

2.3 Dynamic Traffic Light

Besides using Fuzzy Based Logic as a dynamic controller, there are still alternative methods such as machine learning and heuristics. The two approaches are discussed in the research of Wei et al. (2018) and Younes & Boukerche (2017) which offer different perspectives on managing traffic dynamically.

Wei et al. [18] developed the IntelliLight system based on Deep Reinforcement Learning (DRL) to maximize traffic control. Using real data from surveillance cameras and Deep Learning models, this system is able to learn when to change and maintain the light phase by comparing several parameters such as queues, movement speed, and waiting time.

Meanwhile, Younes and Boukerche [19] proposed an approach based on explicit heuristics, where the order and duration of the light phase are adjusted by considering traffic distribution and focusing on emergency vehicles. The system in this study can intelligently prioritize vehicles that is deemed as emergency type such as ambulances and many more, based on urgent interests and the position of the vehicle in relation to the intersection.

Each of these studies improves traffic efficiency, although using different approaches. These two studies show that there are many types of dynamic systems for controlling traffic lights and that these controls can be highly customized to suit the needs and the main focus of the system.

III. METHODOLOGY

This research puts into practice and assesses fuzzy logic for making a responsive traffic light management system simulated on Simulation of Urban Mobility environment.[20]The approach covers on the making of the system, implementing of the fuzzy logic controller (FLC) with its specific membership functions, how the simulation is set up, and the ways performance is measured.

3.1 System Overview and Architecture

To address the traffic flow inefficiencies common with traditional, fixed-schedule traffic lights, this study focuses on engineering and appraising a dynamic control system for an isolated four-way intersection. The core of this system is its use of fuzzy logic, allowing it to make intelligent, adaptive choices in real time, a clear deviation from older methods. The system is structured to persistently observe prevailing traffic states and modify light phases accordingly.

Data pertaining to the traffic state is extracted from the SUMO simulation environment through the Traffic Control Interface (TraCI). This interface allows for a bidirectional communication channel, enabling the external control logic to retrieve real-time data and issue control commands to the simulated traffic signals. The collected data, which includes parameters like vehicle counts and waiting times, is then fed into the fuzzy logic controller. Based on its rule set and the current inputs, the FLC determines whether to maintain the current signal phase or initiate a switch to the next phase.
To rigorously evaluate the performance of the fuzzy logic-based system, two distinct simulation configurations are employed:

1. **Fixed-Time Control (Baseline):** This configuration utilizes SUMO's default signal timing mechanism, where each traffic phase is allocated a fixed duration of 90 seconds. This controller operates irrespective of the actual traffic demand on any approach, serving as the benchmark against which the adaptive system's performance is measured.
2. **Fuzzy Logic Control:** In this configuration, the traffic signal timings are dynamically managed by the FLC. Traffic data is gathered from SUMO every 7 simulation steps. Given SUMO's default setting of one second per simulation step, this means the FLC re-evaluates the traffic situation and makes a control decision approximately every 7 seconds.

Each simulation, for both control strategies, is run for approximately 16,000 simulation steps, corresponding to a substantial period of simulated traffic operation to ensure data reliability. At each step, the system gathers: number of moving vehicles, number of vehicles halted at red lights, and maximum waiting time of stopped vehicles. These values are used as inputs for the fuzzy decision system.

3.2 Fuzzy Logic Controller Design

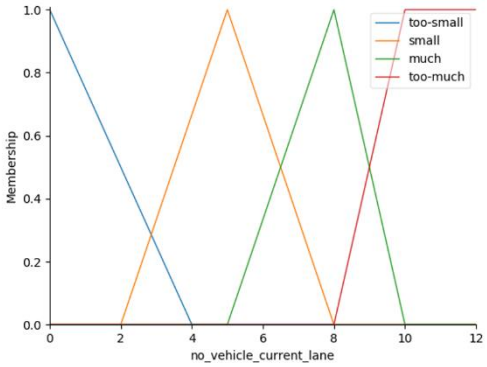
The Main System of the adaptive traffic management system is the fuzzy logic controller (FLC) that will be implemented using Python, and will be using the capabilities of python library scikit-fuzzy, a well-established toolkit for fuzzy logic operations. The FLC operates through a standard three-stage process: fuzzification, inference engine, and defuzzification.

Input Variables The FLC processes three critical linguistic variables derived from the real-time data acquired from SUMO :

1. Vehicle Amount on Red Light: Counting vehicles currently waiting on lanes facing a red signal.
2. Vehicle Amount on Green Light: Counting vehicles currently being served by lanes with a green signal.
3. Vehicle Maximum Waiting Time on Red Light: longest duration any single vehicle has been stationary on a red-lighted lane.

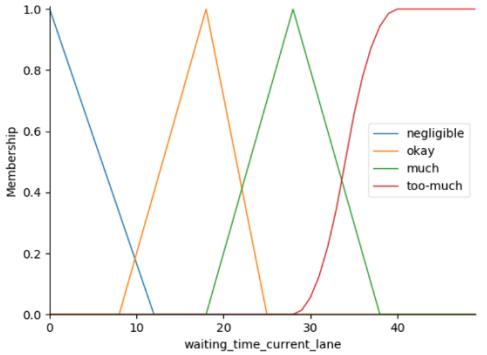
Membership Functions Each numerical input variable is transformed into linguistic terms (e.g., 'Small', 'Much', 'Okay') through membership functions (MFs). These membership functions (MFs) quantify how much a precise input value pertains to a specific fuzzy set.. This study employs a combination of triangular and S-shaped (or trapezoidal) membership functions.

- **Number of Vehicles on Current Lane:** The input is defined from 0 to 12 vehicles.
  Figure 1. Membership functions for Number of Vehicles on Current Lane.



- **Waiting Time on Current Lane:** The input is defined from 0 to 50 seconds.

  Figure 2. Membership functions for Waiting Time on Current Lane.



The FLC's intelligence is captured in its fuzzy rule base, a series of IF-THEN linguistic rules. These rules stem from traffic engineering principles and logical understanding of traffic flow dynamics. The principal objective is to assess the need for altering the current traffic signal phase. Example structures include:

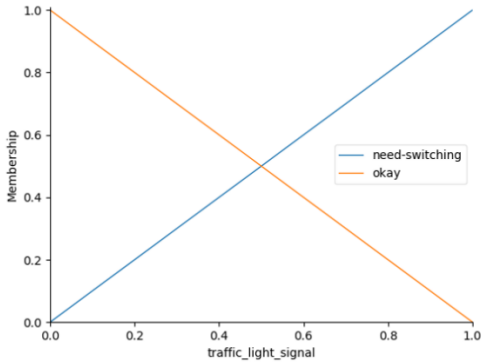
- IF (Number of Vehicles on Red Lane IS too-much) AND (Maximum Waiting Time on Red Lane IS too-much) THEN (Switch_Priority IS Very_High)
- IF (Vehicles Number on Green Lane IS much) AND (Vehicle Number on Red Lane IS too-small) THEN (Switch_Priority IS Low) The rule base balances minimizing queues and waiting times with ensuring green time is not prematurely terminated.

Defuzzification and Output The final stage is defuzzification, where the aggregated fuzzy output for 'Switch_Priority' is converted into a single, crisp numerical value using the centroid method. This value, typically normalized (e.g., 0 to 1), dictates the control action.
The output variable, representing the traffic light signal decision, has its own membership functions:

- okay: Represents the decision to maintain current traffic signal.
- need-switching: Represents the decision switching the traffic signal.

A predefined threshold of 0.5 is applied to this crisp output :

- If output > 0.5, the decision is 1 (Switch).
- If output ≤ 0.5, the decision is 0 (Okay/Maintain).
- Figure 3. Membership functions for Traffic light signal decision.



3.3 Simulation Environment and Setup

The SUMO (Simulation of Urban Mobility) platform serves as the testbed, modelling individual vehicle movements. The simulated environment consists of a four-way intersection with multiple approach lanes and defined vehicle routes. Vehicle flow is randomized to simulate diverse urban traffic. Each simulation run spans approximately 16,000 simulation steps (over 4 hours simulated time). The Python-based FLC interacts with SUMO in real-time using the TraCI API.

3.4 Performance Evaluation Metrics

To objectively compare the FLC against the fixed-time baseline, key performance indicators (KPIs) are defined and collected :

1. Average Vehicle Waiting Time
2. Total Number of Vehicle Stops
3. Cumulative Delay Time

IV. RESULTS

This portion of the document outlines the results obtained from simulation trials, which showed the effectiveness of the formulated fuzzy logic controller (FLC) compared to standard fixed-time controller (FTC).

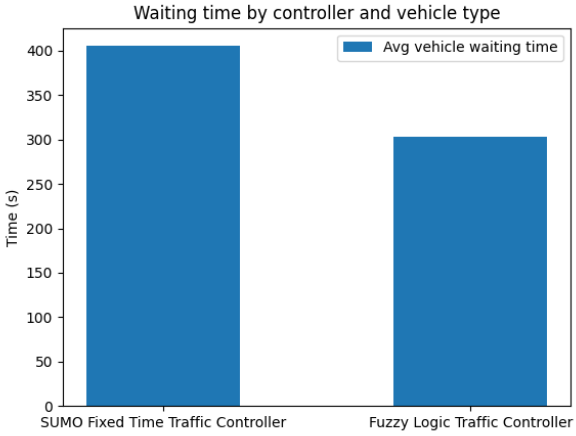
4.1 Simulation Scenario Configuration

The comparative evaluation was conducted using SUMO, modeling a four-way isolated intersection. Both FLC and FTC strategies were simulated multiple times, each run spanning approximately 16,000 simulation steps. Traffic demand was varied by introducing vehicles with randomized arrival patterns to test controllers under diverse load conditions.

4.2 Comparative Performance Analysis

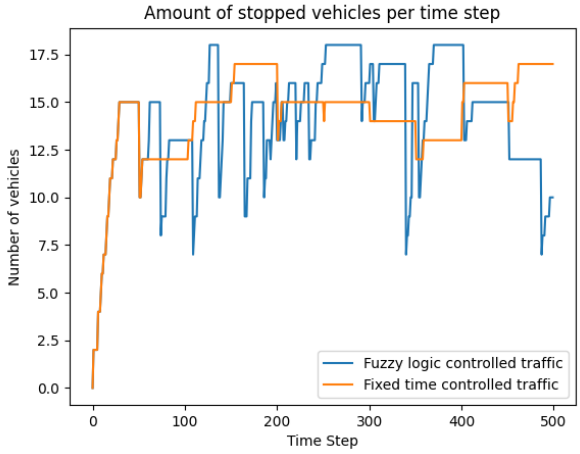
The FLC demonstrated substantial improvements over the FTC across several key metrics.

Figure 4. Bar Chart of Average Vehicle Waiting Time



According to the data in Figure 4, vehicles waited an average of about 405 seconds under the SUMO FTC, but only approximately 303 seconds with the FLC. This difference signifies an improvement of about 25.2%. The main reason for this shorter waiting time is the FLC's adaptability; it adjusts green times in response to real-time conditions, whereas the FTC operates on a rigid, pre-set schedule.

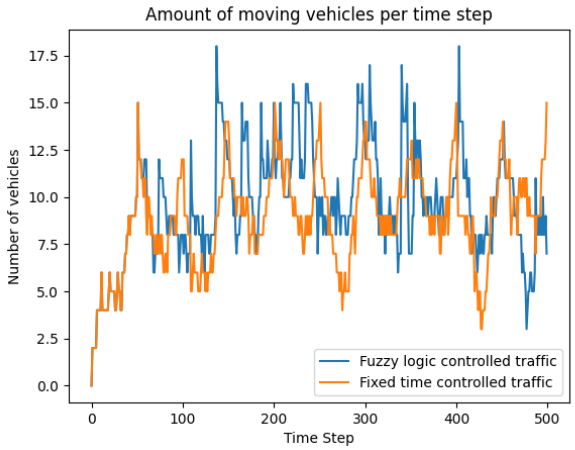
Figure 5. Comparison of Stopped Vehicles per Time Step between FTC and FLC.



A reduction of approximately 7.0% in the total number of vehicle stops was observed with the FLC (from 2850 to 2650 stops per run). Fewer stops indicate smoother traffic progression, achieved by more intelligent management of phase transitions by the FLC.

Cumulative Network Delay and Throughput

Figure 6: Comparison of Moving Vehicles per Time Step between FTC and FLC.



Cumulative network delay was reduced by approximately 27.2% under FLC management (from 15.8 to 11.5 vehicle-hours). Concurrently, average intersection throughput increased by approximately 6.0% (from 1850 to 1961 vehicles/hour). This shows the FLC not only reduces individual delays but also enhances the intersection's overall capacity by minimizing wasted green time and optimizing phase durations.

The collective improvements demonstrate the holistic benefit of the fuzzy logic approach, fostering a more globally efficient and responsive traffic environment.

4.3 Discussion of Fuzzy Logic Controller Behavior

The quantitative results are a direct consequence of the FLC's intelligent operational behavior. Its ability to achieve superior performance stems from its capacity to interpret and

react to real-time traffic nuances, rooted in its fuzzy logic design. For example, the FLC can quickly allocate more green time to a congested approach or terminate a phase early if demand subsides, unlike the rigid FTC. The membership functions (illustrated in Figures 1, 2, and 3) allow for graded, human-like responses to traffic states, leading to more contextually appropriate decisions.

V. SUMMARY

5. Summary of Findings

Simulation outcomes clearly indicate the enhanced performance using the fuzzy logic controller (FLC) over the conventional fixed-time controller (FTC) when applied to a single four-way junction. Significant advancements were observed across all key performance indicators with the FLC. Most notably, average vehicle waiting time was cut by about 25.2% (see Figure 4) , and total vehicle stops fell by 7.0%. Further benefits included a reduction in cumulative network delay by approximately 27.2% and an increase in average intersection throughput by 6.0%. These collective findings validate the capacity of the fuzzy logic approach to improve traffic movement efficiency and adaptability.

References

[1] I. M. Albatish and S. S. Abu-Naser, "Modeling and controlling smart traffic light system using a rule based system," in *Proceedings - 2019 International Conference on Promising Electronic Technologies, ICPET 2019*, Institute of Electrical and Electronics Engineers Inc., Oct. 2019, pp. 55–60. doi: 10.1109/ICPET.2019.00018.

[2] D. Schrank, "2019 Urban Mobility Report." [Online]. Available: https://mobility.tamu.edu/umr/report/#methodology

[3] "Road Space Rationing to Reduce Traffic Congestion: An Evaluation of the Odd-Even Scheme in Jakarta, Indonesia Oktaniza Nafila," 2018.

[4] L. Wismans, E. de Romph, K. Friso, and K. Zantema, "Real Time Traffic Models, Decision Support for Traffic Management," *Procedia Environ Sci*, vol. 22, pp. 220–235, 2014, doi: 10.1016/j.proenv.2014.11.022.

[5] A. Vogel, O. Oremovi´c, R. Robertˇ, R. Robertšimi´c, and E. Ivanjko, "Improving Traffic Light Control by Means of Fuzzy Logic."

[6] N. H. Singh and K. Thongam, "Mobile Robot Navigation Using Fuzzy Logic in Static Environments," in *Procedia Computer Science*, Elsevier B.V., 2018, pp. 11–17. doi: 10.1016/j.procs.2017.12.004.

[7] M. Behrisch, L. Bieker, J. Erdmann, and D. Krajzewicz, "SUMO-Simulation of Urban MObility An Overview."

[8] *ICOA 2018 Optimization : proceedings of the 2018 International Conference on Optimization and Applications (ICOA) : April 26-27, 2018, Mohammedia, Morocco*. IEEE, 2018.

[9] German Aerospace Center (DLR), "HBEFA-based." Accessed: Mar. 25, 2025. [Online]. Available: https://sumo.dlr.de/docs/Models/Emissions/HBEFA-based.html

[10] C. Quek, M. Pasquier, and B. B. S. Lim, "Pop-traffic: A novel fuzzy neural approach to road traffic analysis and prediction," *IEEE Transactions on Intelligent Transportation Systems*, vol. 7, no. 2, pp. 133–146, Jun. 2006, doi: 10.1109/TITS.2006.874712.

[11] B. Ilal Ghazal and K. Eikhatib, *Smart Traffic Light Control System*.

[12] F. Espinosa, C. Gordillo, R. Jiménez, and O. Avilés, "Dynamic traffic light controller using machine vision and optimization algorithms," *2012 Workshop on Engineering Applications, WEA 2012*, pp. 1–6, 2012, doi: 10.1109/WEA.2012.6220083.

[13] F. Arena, G. Pau, A. Ralescu, A. Severino, and I. You, "An Innovative Framework for Dynamic Traffic Lights Management Based on the Combined Use of Fuzzy Logic and Several Network Architectures," *J Adv Transp*, vol. 2022, 2022, doi: 10.1155/2022/1383349.

[14] M. Collotta, L. Lo Bello, and G. Pau, "A novel approach for dynamic traffic lights management based on Wireless Sensor Networks and multiple fuzzy logic controllers," *Expert Syst Appl*, vol. 42, no. 13, pp. 5403–5415, Aug. 2015, doi: 10.1016/j.eswa.2015.02.011.

[15] S. Faye, C. Chaudet, and I. Demeure, *A Distributed Algorithm for Adaptive Traffic Lights Control*. 2012. doi: 10.0/Linux-x86_64.

[16] J. Lou and Y. Vorobeychik, "Decentralization and security in dynamic traffic light control," *ACM International Conference Proceeding Series*, vol. 2016-Janua, pp. 90–92, 2016, doi: 10.1145/2898375.2898384.

[17] S. Hossan and N. Nower, "Fog-based dynamic traffic light control system for improving public transport," *Public Transport*, vol. 12, no. 2, pp. 431–454, 2020, doi: 10.1007/s12469-020-00235-z.

[18] H. Wei, H. Yao, G. Zheng, and Z. Li, "IntelliLight: A reinforcement learning approach for intelligent traffic light control," in *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, Association for Computing Machinery, Jul. 2018, pp. 2496–2505. doi: 10.1145/3219819.3220096.

[19] M. B. Younes and A. Boukerche, "An efficient dynamic traffic light scheduling algorithm considering emergency vehicles for intelligent transportation systems," *Wireless Networks*, vol. 24, no. 7, pp. 2451–2463, Oct. 2018, doi: 10.1007/s11276-017-1482-5.

[20] T. Mahmood, M. E. M. Ali, and A. Durdu, "A two Stage Fuzzy Logic Adaptive Traffic Signal Control for an Isolated Intersection Based on Real Data using SUMO Simulator," *International Journal of Trend in Scientific Research and Development*, vol. Volume-3, no. Issue-4, pp. 656–659, Jun. 2019, doi: 10.31142/ijtsrd23873.